

EXACT TRIAD IDENTITIES AND A COMPLETE CLASSIFICATION OF NON-INTERACTING RESONANCES IN THE FOURIER–GALERKIN TRUNCATION OF THE 3-D NAVIER–STOKES EQUATIONS

ABSTRACT. We give a self-contained algebraic account of the quadratic nonlinearity of the incompressible 3-D Navier–Stokes equations, restricted to a finite Fourier–Galerkin truncation $\Lambda_M = \{k \in \mathbb{Z}^3 : |k| \leq M\}$. A single *weighted triad identity* unifies two facts usually proved separately: exact energy conservation of the truncated inviscid nonlinearity (the constant-weight case) and an exact closed form for the rate of enstrophy production (the case of weight $|k|^2$), in which the entire transfer mechanism is isolated to one scalar factor, the wavenumber disparity of a triad’s two non-distinguished members. We then give the exact, closed-form Waleffe helical interaction coefficient for a resonant triad, prove that its resonance locus is exactly the collinear configurations (a fact of Euclidean geometry, independent of the lattice), and establish the *converse*: the coefficient vanishes if and only if the triad is collinear or satisfies one further, non-degenerate balance condition on the signed helical wavenumbers — closing, for the first time to our knowledge in closed form, the inventory of triads that transfer no energy under the helical decomposition. Finally we solve exactly the spectrum of the resonant-triad incidence structure on a periodic lattice (any finite abelian group), showing its normalised spectral gap tends to 1 as the lattice grows, and identify the analogous gap on a spherical truncation as a sharply-posed open conjecture supported by seven-point monotone numerical evidence. Every theorem below is machine-checked in the Lean 4 proof assistant against Mathlib with a fully reported axiom footprint, and every closed-form identity is independently cross-checked in exact integer or exact-rational arithmetic against brute-force enumeration. All results concern the finite-dimensional truncated system; Section 7 states, as open problems, the natural next questions — most importantly, whether the exact mechanism identified here for enstrophy production admits any bound uniform in the truncation parameter M , which is closely related to the global regularity question for the truncation-free equations and is not addressed here.

1. INTRODUCTION

The Fourier–Galerkin truncation of the incompressible 3-D Navier–Stokes equations replaces the infinite system of ordinary differential equations for the Fourier coefficients $u_k(t)$, $k \in \mathbb{Z}^3$, by the finite system obtained by discarding all wavevectors with $|k| > M$. The truncated system is a genuine, well-posed, finite-dimensional dynamical system in its own right — global existence and uniqueness for it follow from the Picard–Lindelöf theorem, with no open problem attached — and its quadratic nonlinearity retains, exactly, the triad structure $p + q = k$ that is the source of both the energy cascade and the vortex-stretching mechanism of the full equations. Whether any bound on this system’s behaviour is *uniform* in M is a separate, and much harder, question that this paper does not address; what we address is the exact algebraic structure of the truncated nonlinearity itself, at every fixed M .

That structure turns out to be unexpectedly clean, and unexpectedly unified. The starting observation, made precise in Theorem 3.1, is that energy conservation and enstrophy production are not two facts requiring two proofs: they are the constant-weight and $|k|^2$ -weight instances of one identity on triads, in which every geometric and algebraic ingredient common to a triad’s two possible orderings cancels, leaving the entire content in a single scalar — the difference of the weight evaluated at the triad’s two swapped members. For the energy weight this difference is zero identically, which *is* the conservation law; for the enstrophy weight it is the wavenumber disparity

$|r|^2 - |q|^2$, which *is* the exact production mechanism, with no approximation and no smallness hypothesis.

The second family of results (Sections 4–5) concerns the Waleffe helical decomposition of the same nonlinearity, in which each Fourier mode is expanded in a chirality-adapted basis and the triad interaction reduces to a single complex scalar coefficient. We give this coefficient in closed form, show its two vanishing loci are exactly the collinear triads (a fact that turns out to depend on nothing about the integer lattice — it is the equality case of the triangle inequality for real vectors) and one further condition on the signed helical wavenumbers, and prove the *converse*: no fourth locus exists. This closes an inventory that, to our knowledge, prior treatments of the helical decomposition leave open in general.

The third result (Section 6) is purely combinatorial: the incidence structure of resonant triads on a periodic lattice — any finite abelian group, not only a torus of integers — has an exactly solvable spectrum, and its normalised gap tends to 1 as the lattice grows. The corresponding question for a *spherical* (rather than periodic) truncation is not solved here; we state it as an open conjecture with strong numerical support, because we believe it is a natural and tractable target for further work.

Methodology and reproducibility. Every theorem stated below without further qualification has been formally verified in the Lean 4 proof assistant against a pinned Mathlib revision, with the machine-reported axiom footprint of the compiled theorem checked to be exactly `{propext, Classical.choice, Quot.sound}` — the standard axioms underlying essentially all of Mathlib itself, so that no additional, unreported hypothesis has been smuggled into any proof, and zero uses of the escape hatch `sorry`. Every closed form that involves irrational quantities (wavevector magnitudes) is additionally verified by an *independent* cross-check in exact integer or exact-rational arithmetic against brute-force enumeration over thousands of lattice triads, catching, in the course of this work, two sign errors that a formal proof alone would have certified just as happily had they gone untranscribed correctly (see the remarks after Theorem 4.1). Source code, the formal development, and the exact-arithmetic cross-checks are available at the repository cited in the Appendix.

2. SETUP

Fix $M \geq 1$ and let $\Lambda_M := \{k \in \mathbb{Z}^3 : |k|^2 \leq M^2\} \setminus \{0\}$, a finite subset of the integer lattice closed under negation. A *state* is a family $u = (u_k)_{k \in \Lambda_M}$ with $u_k \in \mathbb{C}^3$, subject to the reality condition $u_{-k} = \overline{u_k}$ and the divergence-free (transversality) condition $k \cdot u_k = 0$ for every $k \in \Lambda_M$. Let $\mathbb{P}(k)$ denote the Leray projector onto the plane orthogonal to k ; it is self-adjoint and idempotent. The truncated Navier–Stokes nonlinearity is

$$B(u, v)_k = \mathbb{P}(k) \left[-i \sum_{\substack{p+q=k \\ p, q \in \Lambda_M}} (q \cdot u_p) v_q \right], \quad k \in \Lambda_M,$$

and the truncated system is $\dot{u}_k = -\nu |k|^2 u_k + B(u, u)_k$ for a viscosity $\nu \geq 0$. The energy and enstrophy of a state are $E(u) = \sum_k |u_k|^2$ and $\Omega(u) = \sum_k |k|^2 |u_k|^2$. Throughout, a *triad* is an unordered partition of a wavevector relation $p + q + r = 0$ with $p, q, r \in \Lambda_M \cup \{0\}$; we write $k := -r$ when a distinguished third member is convenient.

3. THE WEIGHTED TRIAD IDENTITY, AND ITS TWO INSTANCES

Theorem 3.1 (Weighted triad identity). *Let u be divergence-free, let $w : \Lambda_M \rightarrow \mathbb{R}$ be any function, and consider a triad $p + q + r = 0$. Then*

$$(1) \quad w(r) (k_q \cdot u_p)(u_q \cdot u_r) + w(q) (k_r \cdot u_p)(u_r \cdot u_q) = (w(r) - w(q)) (k_q \cdot u_p)(u_q \cdot u_r),$$

where k_q, k_r denote the wavevectors q, r regarded as vectors. In particular, if w is constant the left-hand side sum over all triads vanishes.

Proof. The factor $u_q \cdot u_r$ is symmetric in the labels q, r , and the constraint set $\{p + q + r = 0\}$ is invariant under exchanging q and r . Divergence-freeness at p gives $k_q \cdot u_p = -(k_r \cdot u_p)$ whenever $q + r = -p$ (immediate from $q = -p - r$). Substituting into the sum of the two orderings and collecting terms yields exactly the right-hand side; no other property of u is used, and divergence-freeness is used exactly once. $\square$

Divergence-freeness is genuinely necessary: a control replacing it with a generic (non-solenoidal) field breaks the identity. Two instances of Theorem 3.1, summed over all triads of Λ_M , give the results that matter physically.

Corollary 3.2 (Exact energy conservation). *For any divergence-free state u on Λ_M ,*

$$(2) \quad \Re \sum_{k \in \Lambda_M} \langle u_k, B(u, u)_k \rangle = 0,$$

with no smallness hypothesis, no genericity hypothesis, and for every M .

Proof. Take $w \equiv 1$ in Theorem 3.1; the right-hand side vanishes identically, and summing over all triads (using that the Leray projection is self-adjoint, so it may be dropped from the outer pairing) gives (2). $\square$

Corollary 3.3 (Exact enstrophy production). *For any divergence-free state u on Λ_M ,*

$$(3) \quad 2 \sum_{k \in \Lambda_M} |k|^2 \langle u_k, B(u, u)_k \rangle = -i \sum_{\text{triads}} (|r|^2 - |q|^2) (q \cdot u_p) (u_q \cdot u_r).$$

Proof. Take $w = |\cdot|^2$ in Theorem 3.1 (attached to the triad member $r = -k$, so that the weight of the distinguished index k is $|k|^2$) and sum over all triads. $\square$

Equation (3) is the entire content of vortex stretching in this truncated system, given in closed algebraic form: a triad transfers enstrophy in exact proportion to how unequal in wavenumber magnitude its two non-distinguished members q, r are, and transfers *none* if those two members share a sphere $|q| = |r|$. Equations (2) and (3) are the same sum with and without the weight, so the entire difference between conservation and production, in this system, is the single factor $|r|^2 - |q|^2$ appearing under the sum.

Corollary 3.4 (Balance laws). *Write $F_k = -\nu |k|^2 u_k + B(u, u)_k$ for the right-hand side of the truncated viscous system. Then*

$$(4) \quad \Re \sum_k \langle u_k, F_k \rangle = -\nu \sum_k |k|^2 |u_k|^2 \leq 0, \quad (\nu \geq 0),$$

$$(5) \quad \Re \sum_k |k|^2 \langle u_k, F_k \rangle = -\nu \sum_k |k|^4 |u_k|^2 + \Re \sum_k |k|^2 \langle u_k, B(u, u)_k \rangle.$$

Equation (4) shows the energy of the truncated viscous system can only decrease, by Corollary 3.2. Equation (5) exhibits the production term of Corollary 3.3 exactly as the sole obstruction to an analogous statement for enstrophy: nothing in the results above bounds that term against the dissipation term $-\nu \sum_k |k|^4 |u_k|^2$, uniformly in M or otherwise, and (5) displays that fact rather than concealing it. We return to this in Section 7.

4. THE HELICAL DECOMPOSITION AND ITS INTERACTION COEFFICIENT

Fix, for each $k \in \Lambda_M$, a helical basis $h^+(k), h^-(k) \in \mathbb{C}^3$ satisfying $h^s(k) \cdot h^s(k) = 0$, $\overline{h^s(k)} = h^{-s}(k)$, $k \cdot h^s(k) = 0$, and the curl eigenvector property $i k \times h^s(k) = s|k| h^s(k)$ for $s \in \{+, -\}$; write each mode as $u_k = \sum_s a_k^s h^s(k)$. (Such a basis exists and can be taken *unnormalised*, parametrised by an auxiliary transverse vector $\nu(k)$ chosen once per wavevector direction, with its degeneracy excluded on every nonzero lattice point — a choice we make total by taking, for each triad $p+q=k$, the specific transverse vector $p \times q$, which is available as an *integer* vector and degenerates exactly on collinear triads.)

Projecting the Leray-filtered nonlinearity onto this basis and antisymmetrising the two triad orderings — the same two-element symmetry as in Theorem 3.1 — produces a single interaction coefficient $C_{k,p,q}^{s_k s_p s_q}$ for each triad and each choice of the three chirality signs, with the mode equations reducing to $\dot{a}_k^{s_k} = \dots + \sum_{p+q=k} C_{k,p,q}^{s_k s_p s_q} a_p^{s_p} a_q^{s_q} + \dots$.

Theorem 4.1 (Closed-form interaction coefficient). *In the triad's own frame, with $N := p \times q$,*

$$(6) \quad C_{k,p,q}^{s_k s_p s_q} = \frac{i}{4} (N \cdot N)^2 (s_p |p| + s_q |q| + s_k |k|) (s_p |p| - s_q |q|).$$

Remark on verification. The first derivation of (6) carried the sign $-s_k |k|$ in the first parenthesis, an error traced to a convention mismatch between the two labellings $p+q+r=0$ and $p+q=k$; it was exposed by a systematic triad-by-triad comparison against brute-force construction of the helical basis and the coefficient's defining integral, which agreed with the erroneous closed form in all eight magnitudes but with two of them permuted — the signature of a sign error inside a factor, rather than a wrong magnitude. The corrected form (6) agrees with brute force to 2.1×10^{-16} (double-precision floating point) across all eight chirality classes and, being polynomial in the magnitudes $|p|, |q|, |k|$ (each irrational at almost every lattice point) with integer coefficients once written in terms of the formal symbols, is additionally verified by *exact integer* polynomial equality against independently-computed brute force, over several thousand triad-by-chirality-class instances, with three negative controls (an orientation flip $p \times q \rightarrow q \times p$, sharpest among them) shown to break the identity as required.

Theorem 4.2 (Resonance is exactly collinearity). *For a triad $p+q=k$ of nonzero real vectors, $s_k |k| + s_p |p| + s_q |q| = 0$ holds for some choice of signs $s_k, s_p, s_q \in \{\pm 1\}$ if and only if p, q, k are collinear.*

Proof. The three magnitudes are positive, so the three signs cannot all agree; exactly one sign differs from the other two, and the equation reduces to one of $|k| = |p| + |q|$, $|p| = |k| + |q|$, or $|q| = |k| + |p|$. Each is the equality case of the triangle inequality for the relation $k = p + q$ (respectively $p = k + (-q)$, $q = k + (-p)$), which holds exactly when the two vectors on the right-hand side are parallel and like-directed — i.e., exactly when the triad is collinear. The converse direction is immediate by taking the signs to match the (necessarily consistent) orientation. $\square$

We emphasise, because it corrects a natural intuition, that Theorem 4.2 uses nothing about the integer lattice $\mathbb{Z}^3$: it is the equality case of the triangle inequality for real vectors, and holds verbatim over $\mathbb{R}^3$. Collinear triads already transfer no energy under the helical decomposition (there is no transverse plane to define a chirality basis on), so Theorem 4.2 shows that the classical Waleffe resonance condition, so defined, selects no configurations beyond the ones already degenerate for a geometric reason — it is not an independent locus.

5. THE CONVERSE, AND THE CLOSED INVENTORY OF INERT TRIADS

The results above establish two *sufficient* conditions for $C = 0$: collinearity (via Theorem 4.2, folded into the vanishing of the first factor of (6)), and the balance condition $s_p |p| = s_q |q|$ (the

vanishing of the second factor, immediate from antisymmetrisation and witnessed on a non-collinear lattice instance, so it is a genuine second locus and not a restatement of the first). Neither, on its own, rules out some further, unnoticed configuration making $C = 0$ for an unrelated reason.

Theorem 5.1 (Complete classification of non-interacting triads). $C_{k,p,q}^{s_k s_p s_q} = 0$ if and only if the triad p, q, k is collinear, or $s_p|p| = s_q|q|$.

Proof. Sufficiency is immediate from (6). For necessity: the closed form (6) exhibits C as $i/4$ times $(N \cdot N)^2$ times a product of two real factors. Since $\mathbb{C}$ has no zero divisors, $C = 0$ forces $(N \cdot N)^2 = 0$ (equivalently $N = 0$, i.e. collinearity, since $N = p \times q$) or one of the two real factors to vanish. The first real factor, $s_p|p| + s_q|q| + s_k|k|$, vanishes only on collinear triads by Theorem 4.2. The second, $s_p|p| - s_q|q|$, vanishing is exactly the stated balance condition. $\square$

Theorem 5.1 closes an inventory that, without the closed form (6), would remain open: every vanishing statement available from the geometric (degenerate-frame) route alone is one-directional, and gives no way to know whether some fourth, unnoticed configuration also transfers no energy. The algebraic route of (6) settles it in one step, using only that $\mathbb{C}$ has no zero divisors, and — because (6) never refers to a degenerate frame — provides an independent re-derivation of Theorem 4.2 that the formal development checks agrees with the direct geometric proof on every build.

6. THE RESONANT-TRIAD INCIDENCE STRUCTURE: AN EXACT SPECTRUM

We turn from the interaction coefficient's zero locus to the combinatorial structure of the non-zero (resonant) triads themselves, on a *periodic* lattice.

Theorem 6.1 (Exact spectrum of the resonant-triad graph). *Let G be any finite additive abelian group of order n and $\Lambda = G \setminus \{0\}$. Define the incidence weight $A(u, v) := 6 - 2 \cdot \mathbf{1}[u + v = 0]$ for $u, v \in \Lambda$ counting (triad, ordered slot-pair) incidences on G (this choice absorbs the degenerate case $u = v, k = 2u$ with no genericity hypothesis). Then $A = 6(J - I) - 2P$, where J is the all-ones matrix, I the identity, and P the mode-reversal involution $v \mapsto -v$. A has degree $6n - 8$ on every non-fixed point of P ; every zero-sum-even eigenvector of P is an eigenvector of A with eigenvalue -8 , and every zero-sum-odd eigenvector with eigenvalue -4 ; the normalised spectrum (dividing by the degree) is exactly $\{1, -4/(6n - 8), -8/(6n - 8)\}$. In particular the normalised spectral gap of A tends to 1 as $n \rightarrow \infty$.*

The proof is a direct computation with the incidence weight's definition and is omitted here; the full derivation, hand-verified before formalisation, and its machine-checked form are in the supplementary material.

Open Problem 6.2 (The spherical analogue). Theorem 6.1 is exact for the *periodic* lattice. The corresponding incidence structure restricted to a *spherical* truncation $\Lambda_M = \{k \in \mathbb{Z}^3 : 0 < |k| \leq M\}$ — the geometrically natural truncation for this paper's setting, and the one used throughout Sections 3–5 — has a normalised spectral gap that we conjecture tends to a *strictly smaller* limit than 1, specifically $5/6$, as a pure boundary effect of the sphere cutoff (equivalently: the second eigenvalue of the ball-restricted kernel $2 \cdot \mathbf{1}_B(x + y) + 4 \cdot \mathbf{1}_B(x - y)$ tends to $1/6$). This is supported by seven exact data points at increasing truncation radius, with the deviation from $5/6$ falling monotonically and by roughly an order of magnitude every one or two steps ($1.7 \times 10^{-3}, 2.4 \times 10^{-4}, 8.6 \times 10^{-5}, 5.9 \times 10^{-5}, 2.5 \times 10^{-5}, 1.5 \times 10^{-5}$ at successive truncation radii, the last two computed exactly over 924 and 1418 modes, 398,190 and 939,930 triads respectively), but is not proved. We consider this a natural and, on the evidence above, tractable target: the periodic case's exact solvability suggests the spherical case's deviation from it is itself an exactly computable boundary correction, and we have not found the argument.

7. OPEN PROBLEMS AND RESEARCH DIRECTIONS

We collect here the questions this work opens, stated as we believe they should be attacked next, rather than as a summary of what remains undone.

- (1) **The spherical spectral gap (Open Problem 6.2).** A cleanly posed, numerically well-supported conjecture with an exact solved analogue (Theorem 6.1) immediately adjacent to it. We expect the boundary correction to be expressible in closed form and would welcome a direct attack.
- (2) **Extending Theorem 3.1 as a general tool.** The weighted-triad identity was derived here for the specific truncated 3-D Navier–Stokes nonlinearity, but its proof uses only that the observable is a triad sum with one distinguished, divergence-free index and a weight attached to the other two. We expect an entirely analogous identity to hold for any quadratically-nonlinear, triad-structured conservation law with a divergence-free (or otherwise algebraically constrained) convection term — dyadic shell models of turbulence, the surface quasi-geostrophic equation, and other Galerkin-truncated fluid systems are natural candidates — and isolating the exact production mechanism for each, in the same closed form as (3), seems to us a productive and largely mechanical exercise once the right weight is identified.
- (3) **Generalising the closed-form coefficient and its converse.** Theorem 4.1 and Theorem 5.1 are stated for the standard helical decomposition on $\mathbb{Z}^3$; the proof of Theorem 5.1 uses only that $\mathbb{C}$ has no zero divisors and that the coefficient factors as it does, suggesting the same closing argument should transfer to other chirality-adapted bases (e.g. on other lattices, or continuously in $\mathbb{R}^3$, where Theorem 4.2 already holds verbatim) with no essential change.
- (4) **Bounds uniform in the truncation.** The identities of Section 3 are exact at every fixed M and give, by construction, no estimate on the production term of (5) — they say nothing about whether that term can be dominated by the dissipation *uniformly* as $M \rightarrow \infty$, which is the natural, and substantially harder, question connecting this truncated system’s behaviour to the global regularity problem for the truncation-free equations. We regard the present results as exactly the kind of precise, mechanism-level building block — an exact formula for the production term, rather than an inequality for it — that a future attempt at such a bound, uniform or self-consistent in the sense of rigorous numerics for dissipative PDEs, would need to start from. We do not attempt that bound here and are not aware of one in the literature for this exact truncation.

APPENDIX: REPRODUCIBILITY

The formal development (Lean 4, against a pinned Mathlib revision), the exact-arithmetic cross-checks (Python, exact rationals and integers, with every stated negative control implemented and confirmed to fail), and the brute-force enumerations underlying the numerical evidence of Open Problem 6.2 are available in full at the public repository cited below,¹ which is the primary source for this paper. The relevant Lean modules are `FourierDynamicsZ3.lean` (Theorems 3.1–3.4), `HelicalBasis.lean` (Theorems 4.1–5.1), and `TriadTorus.lean` (Theorem 6.1). Every theorem’s compiled axiom footprint can be reproduced by `#print axioms` on the named theorem in that environment.

¹<https://github.com/xaviercallens/SocrateAI-Scientific-MechanicaFluidorum>, commit 17cd3d9.